

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

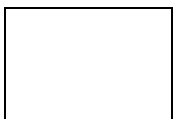
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

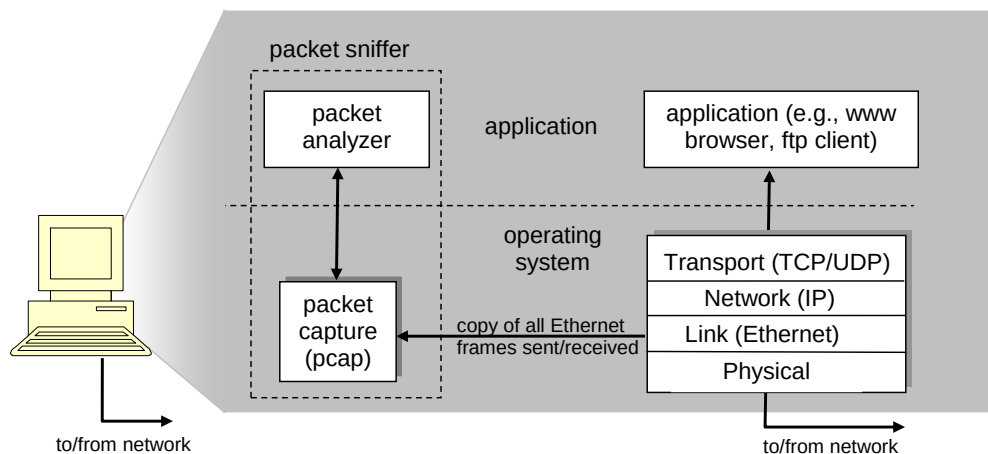


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

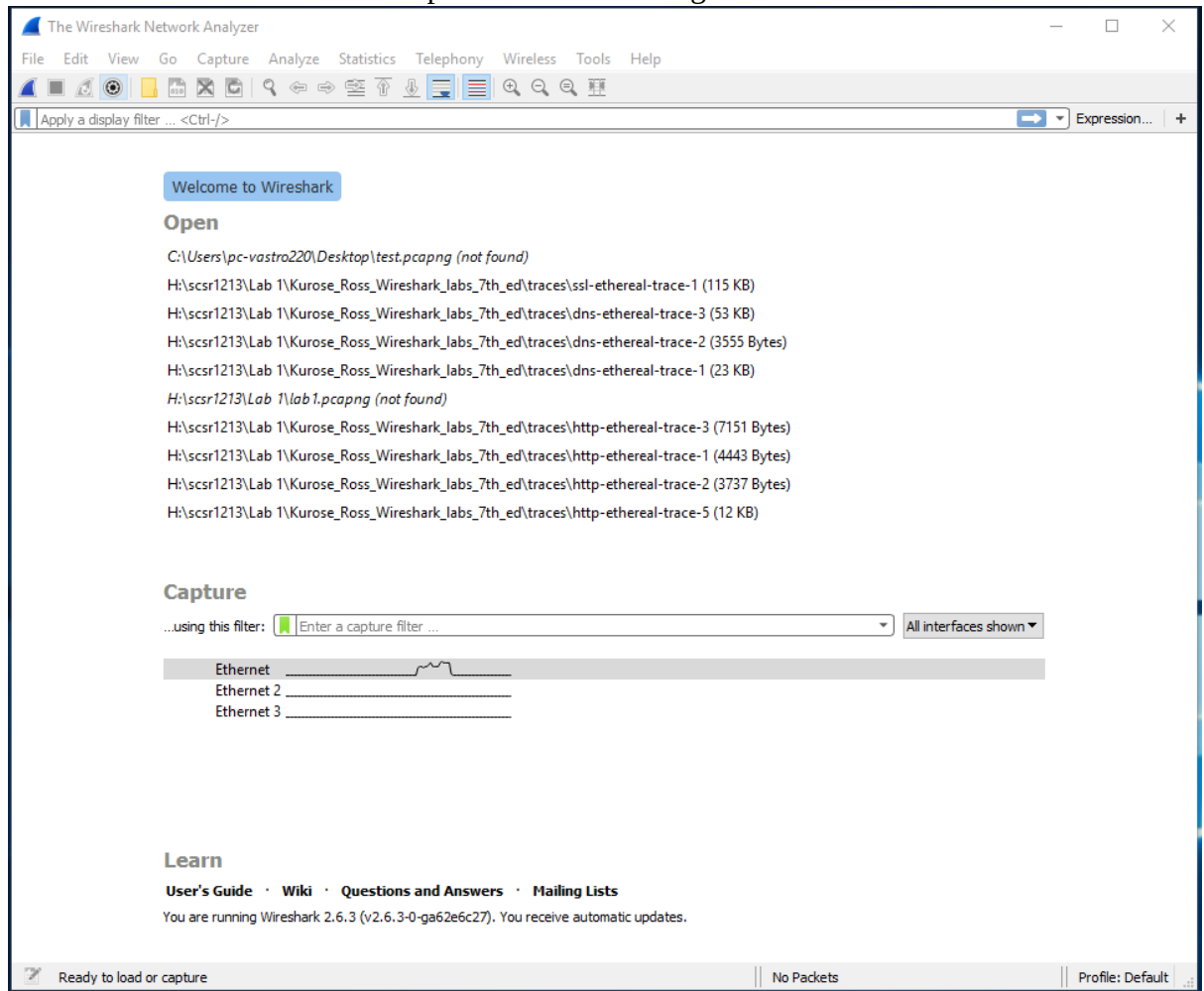


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

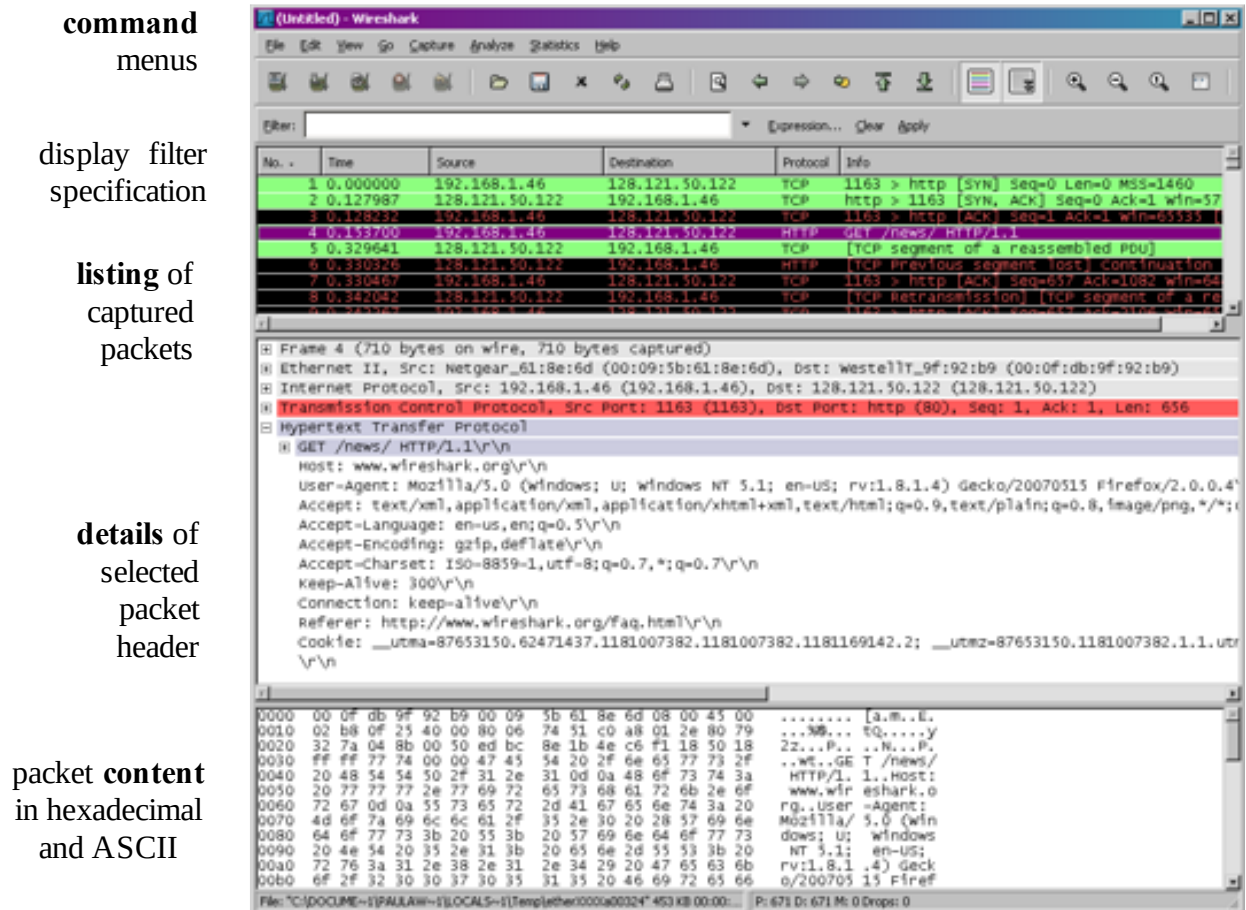


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

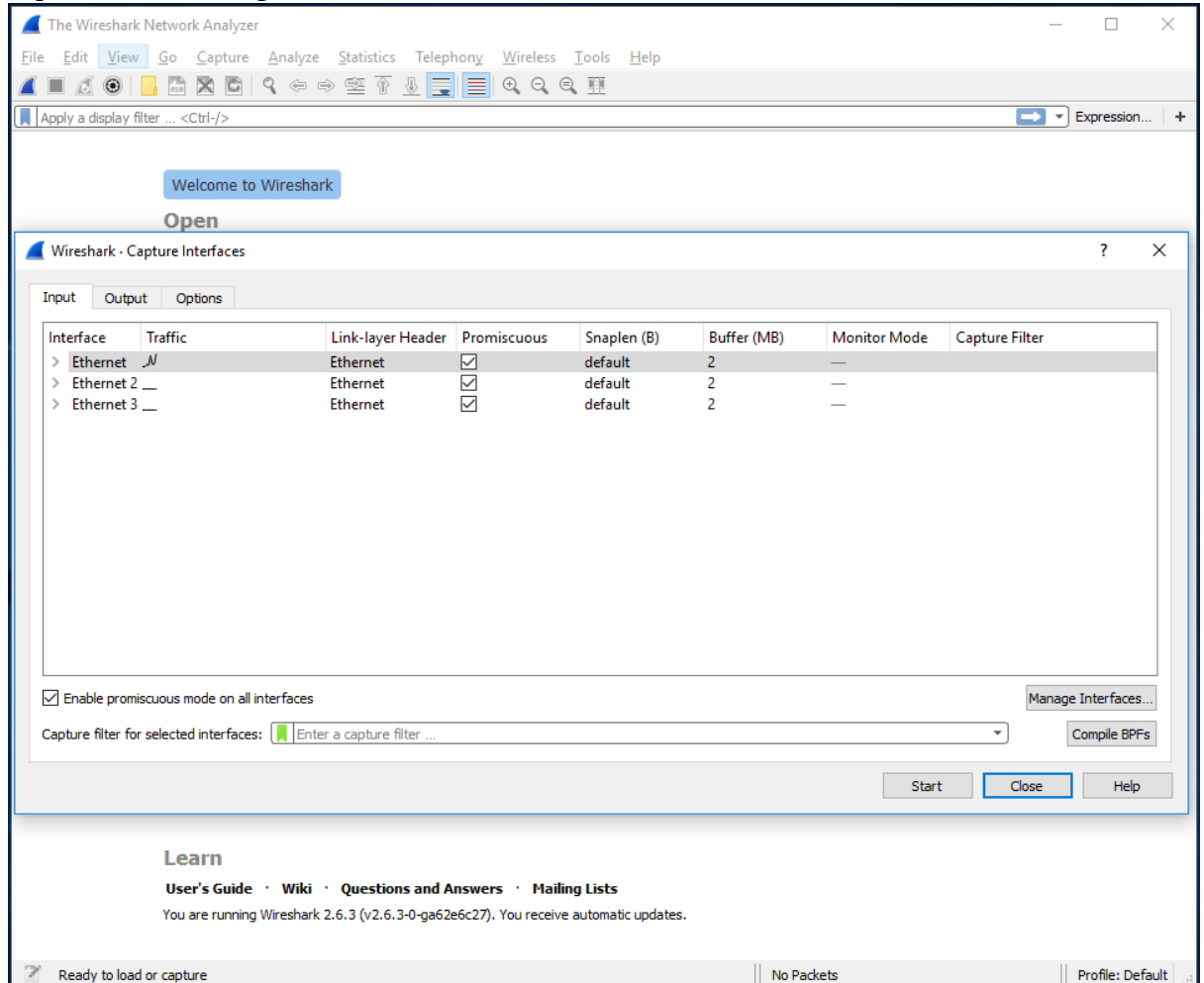


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

The image shows the Wireshark interface with a packet capture in progress. The top pane displays a list of 90 captured packets. The bottom pane shows the details of the selected packet (No. 90), which is an Ethernet II frame from Dell_23:e3:3a to Cisco_d5:79:ff, containing an Internet Protocol Version 4 packet from 10.60.82.216 to 52.230.84.0, and a Transmission Control Protocol packet from port 49724 to port 443.

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
 Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
 Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
 Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
 Secure Sockets Layer

0000 00 14 6a d5 79 ff 00 24 e8 23 e3 3a 08 00 45 00 ..j.y..\$ #.:..E.
 0010 00 5d 64 e9 40 00 80 06 00 00 0a 3c 52 d8 34 e6 ..]d.@...<R.4.
 0020 54 00 c2 3c 01 bb 11 3f 6d 43 3e d4 b9 24 50 18 T...<...? mC>..\$P.
 0030 01 01 e6 49 00 00 17 03 01 00 30 07 75 ee af 3c ...I.....0-u...<
 0040 42 81 47 9e 52 29 c3 be ea 2e 88 ac 28 ed bf e6 B.G.R)...((...
 0050 20 36 00 a7 ad f6 ff 8a b0 6d 63 a3 c7 a4 68 ca 6.....mc...h.
 0060 fd 6d b3 05 36 73 0e 3c 54 bc 07 ..m...6s.<T..

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

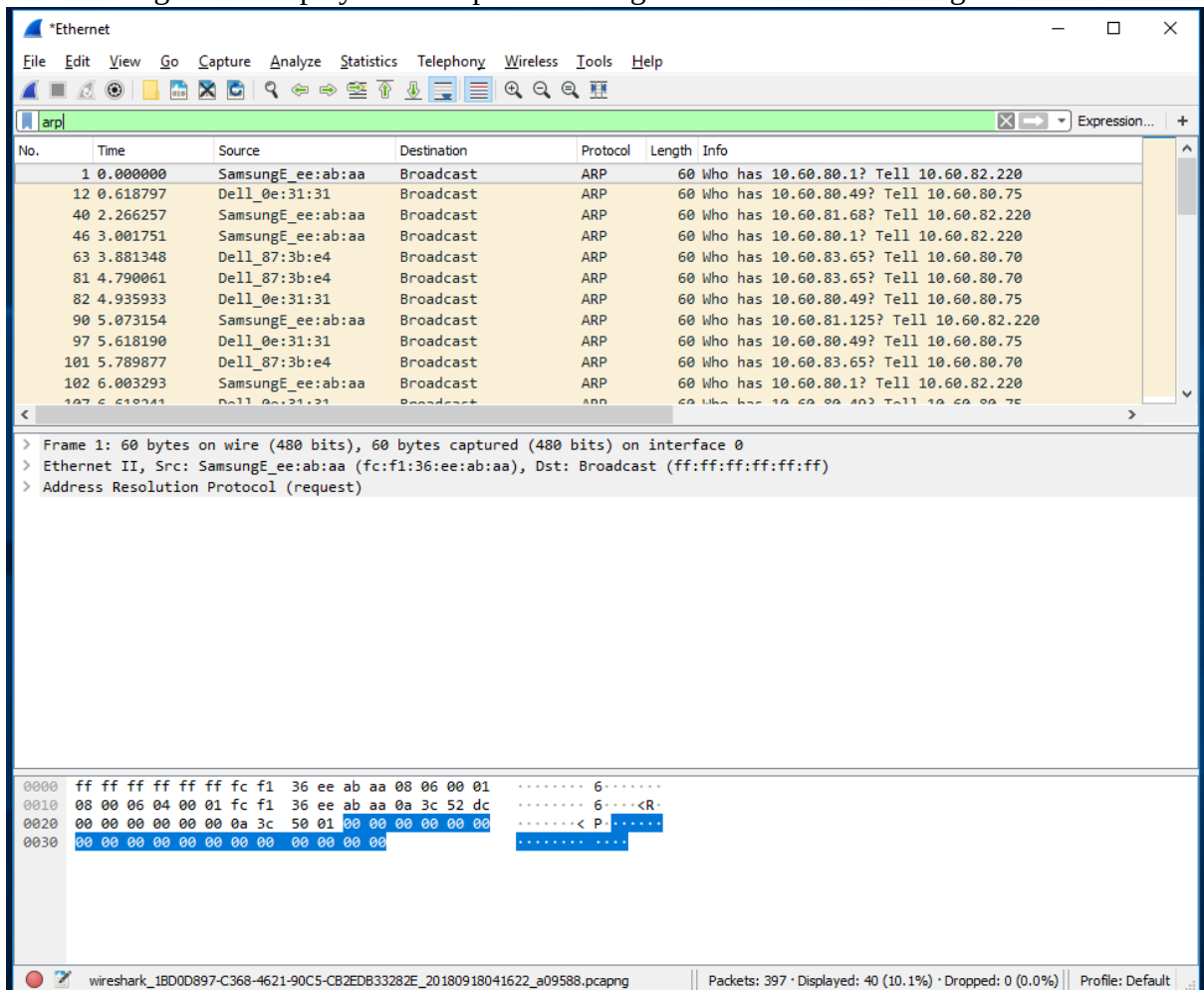


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

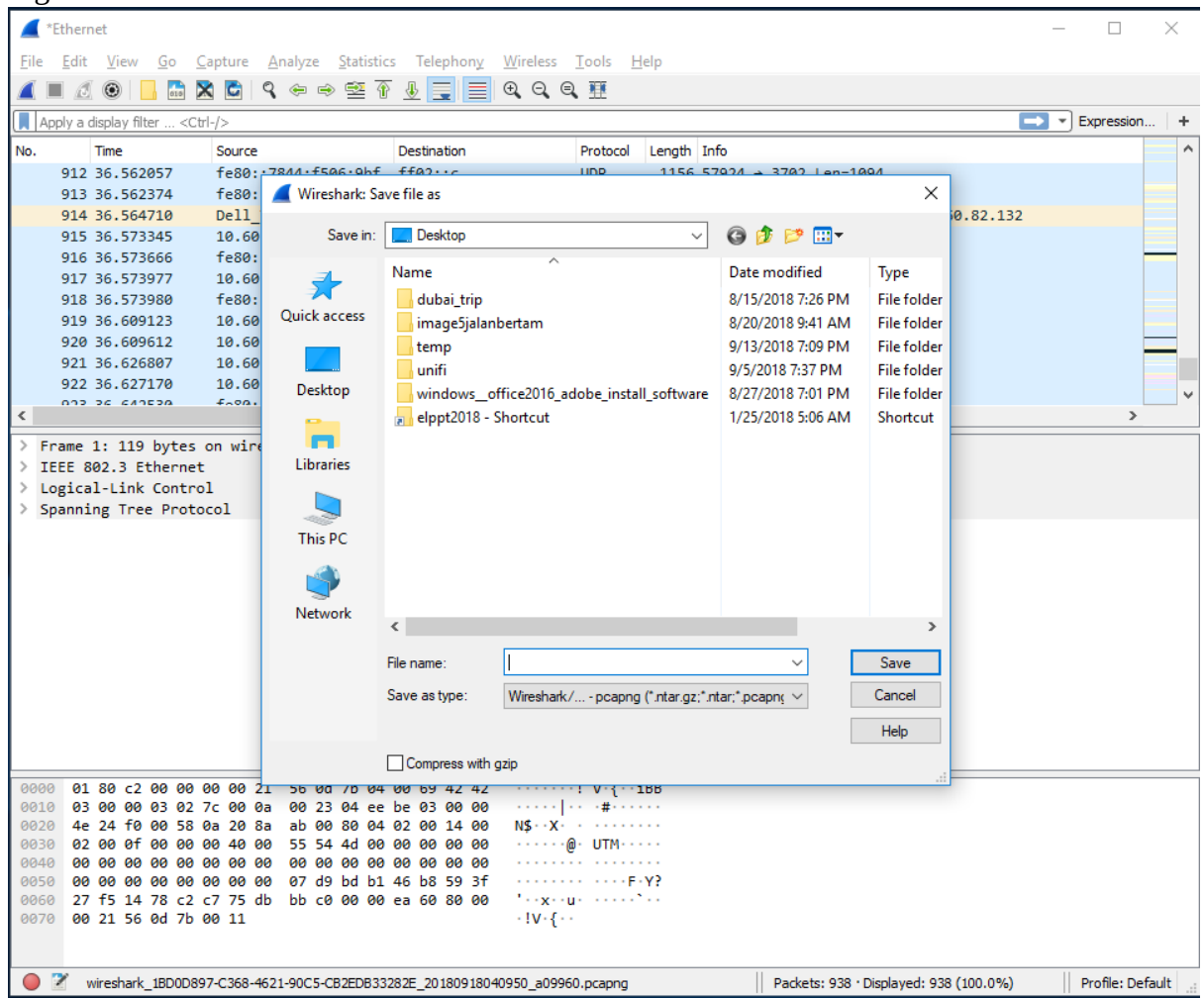
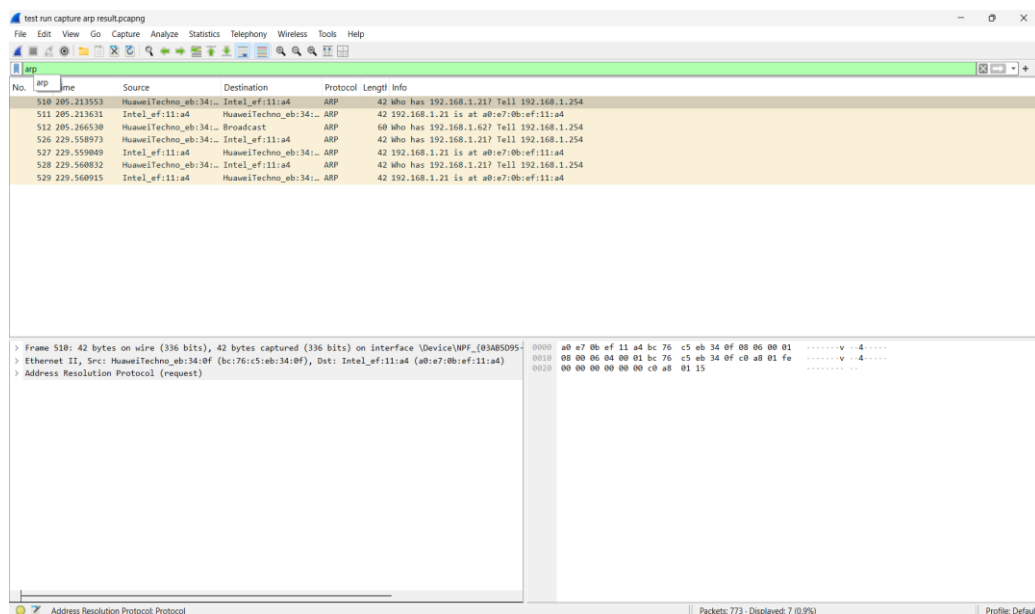


Figure A.7: Save Wireshark trace result

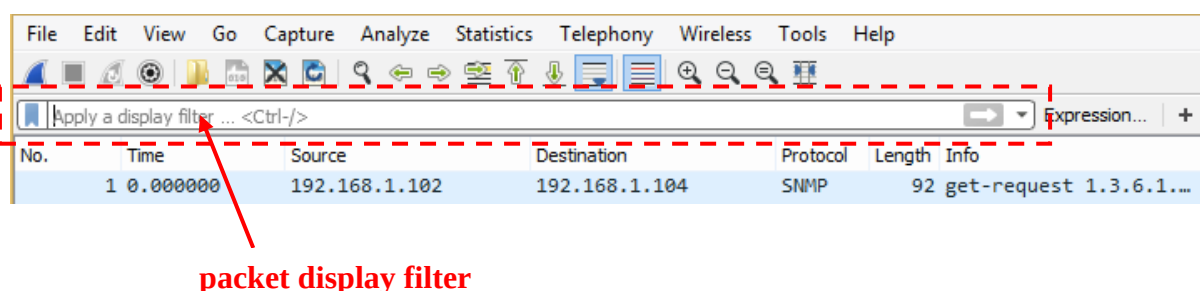


PART B: HTTP Trace

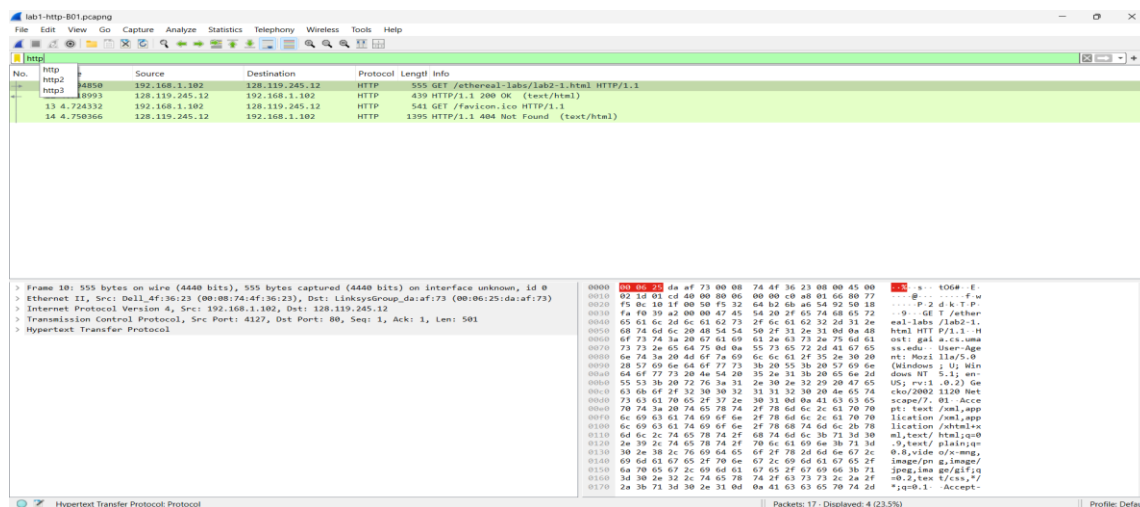
In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter "**http**" (just the letters, not the quotation marks) in the **packet display filter** field, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



- By looking at the information in the HTTP GET and response messages, answer the following questions:



1. What version of HTTP is the server running?
HTTP Version is HTTP/1.1
2. What is the IP address of the client computer?
192.168.1.102
3. What is the IP address of the gaia.cs.umass.edu server?
128.119.245.12

4. How many bytes of content are being returned to client browser?

Frame 12: 439 bytes

Frame 14: 1395 bytes

5. What is the status code returned from the server to client browser?

Frame 12: 200 OK, this indicate that the server has successfully processed the client's request and is returning the requested resource.

Frame 14: 404 Not Found

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server.
Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No, I do not see an “IF-MODIFIED-SINCE” line in the HTTP GET.

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes, the server explicitly returns the contents of the file, because in Frame 10, the server responded a 200 OK status code. It showed the requested file is successfully returned.

The image shows a Wireshark packet capture analysis of the file lab1-http-B02.pcapng. The packet list pane displays four packets:

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	595	GET /etherreal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /etherreal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

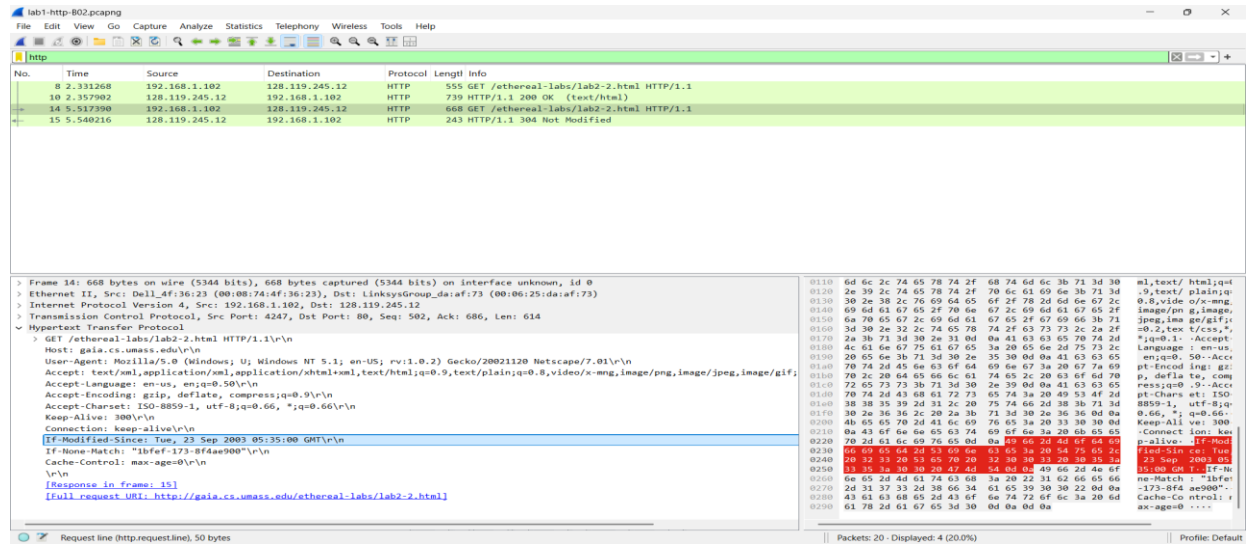
The packet details pane for Frame 10 shows the following information:

- Frame 10: 739 bytes on wire (5912 bits), 739 bytes captured (5912 bits) on interface unknown, id 0
- Ethernet II, Src: LinksysGroup_da:af:73 (00:06:25:da:af:73), Dst: Dell_Af:36:23 (00:08:74:4f:36:23)
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
- Transmission Control Protocol, Src Port: 80, Dst Port: 4247, Seq: 1, Ack: 502, Len: 685
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - Date: Tue, 23 Sep 2003 05:35:50 GMT\r\n
 - Server: Apache/2.0.40 (Red Hat Linux)\r\n
 - Last-Modified: Tue, 23 Sep 2003 05:35:00 GMT\r\n
 - ETag: "1bfef-173-bf4ae900"\r\n
 - Accept-Ranges: bytes\r\n
 - Content-Length: 371\r\n
 - Keep-Alive: timeout=10, max=100\r\n
 - Connection: Keep-Alive\r\n
 - Content-Type: text/html; charset=ISO-8859-1\r\n
 - \r\n
 - [Request in frame: 8]
 - [Time since request: 0.026634000 seconds]
 - [Request URI: /etherreal-labs/lab2-2.html]
 - [Full request URI: http://gaia.cs.umass.edu/etherreal-labs/lab2-2.html]
 - File Data: 371 bytes

The status bar at the bottom indicates: Packets: 20 · Displayed: 4 (20.0%)

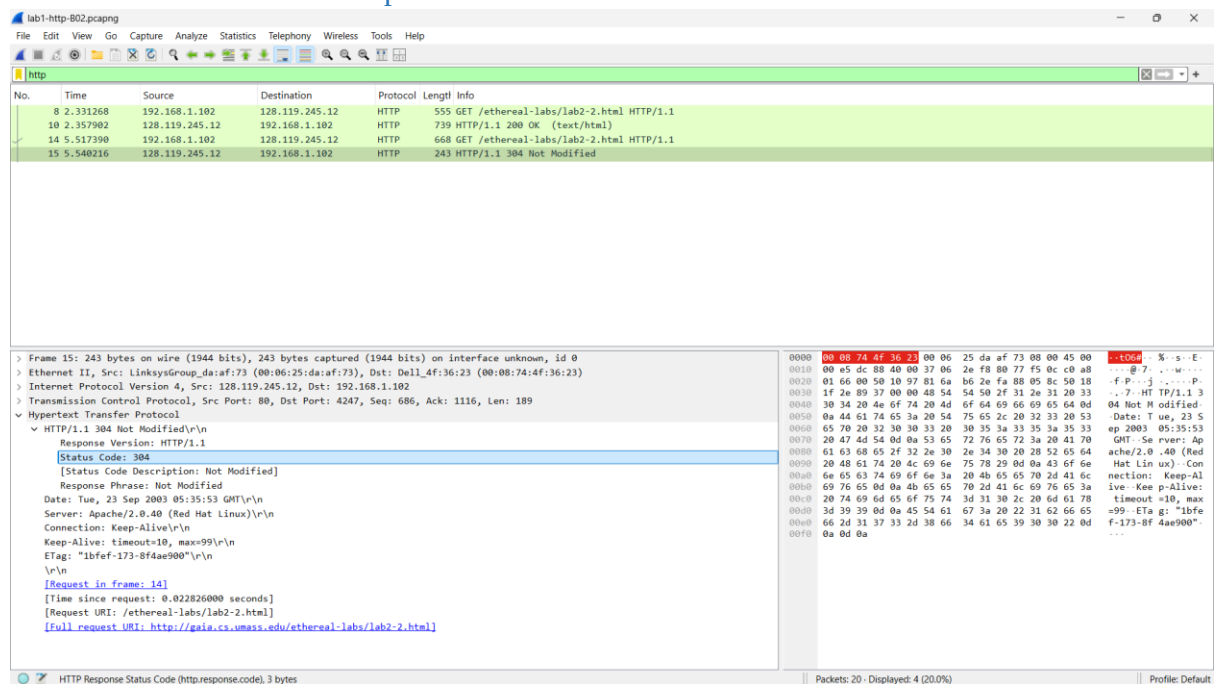
3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, I see an “IF-MODIFIED-SINCE:” line in the HTTP GET, the information follows the “IF-MODIFIED-SINCE:” header is TUE,23 Sep “2003 05:35:00 GMT”, which means if the resource on the server has been modified after the date specified, the server will respond with a 200 OK status and send the updated content back to the browser.



4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The HTTP status code is 304 Not Modified. This indicates that the server did not explicitly return the contents of the file because it determined that the file has not been modified since the date provided in the "IF-MODIFIED-SINCE" header.



B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

The client browser sent 3 HTTP GET request messages.

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

2. To which Internet addresses were these GET requests sent?

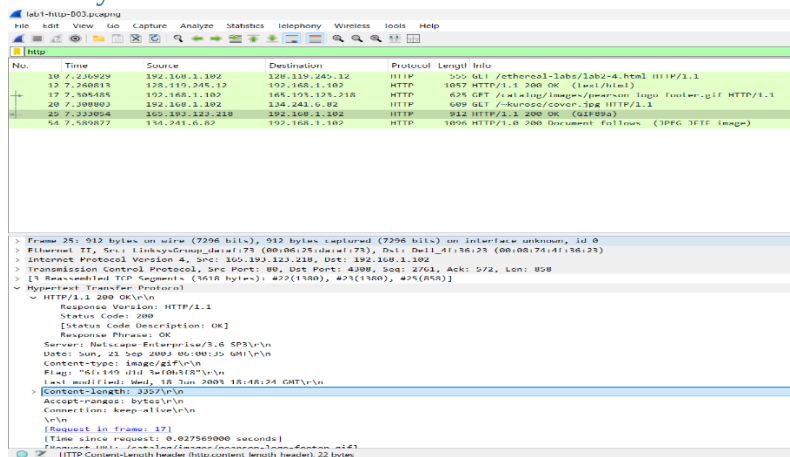
1st request were sent to 128.119.245.12.

2nd request were sent to 165.193.123.218

3rd request were sent to 134.241.6.82

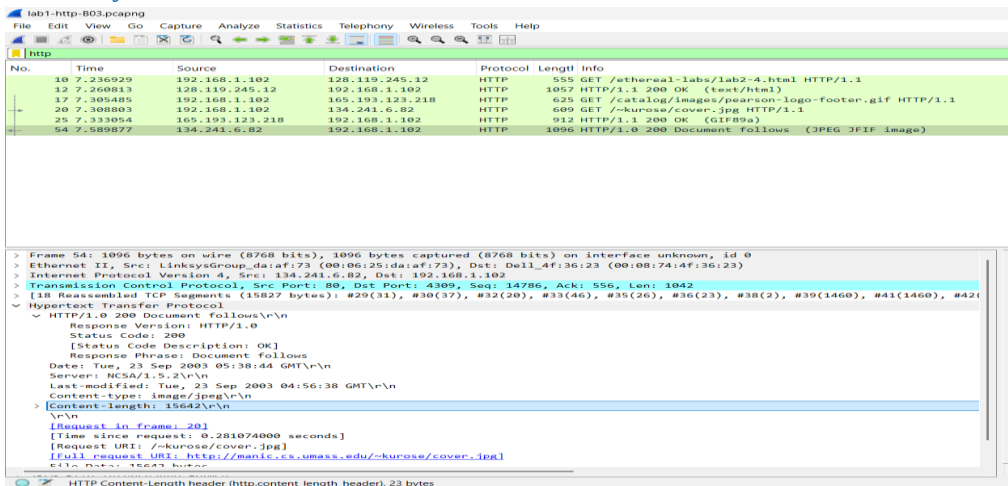
3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

3357 bytes



4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

15642 bytes

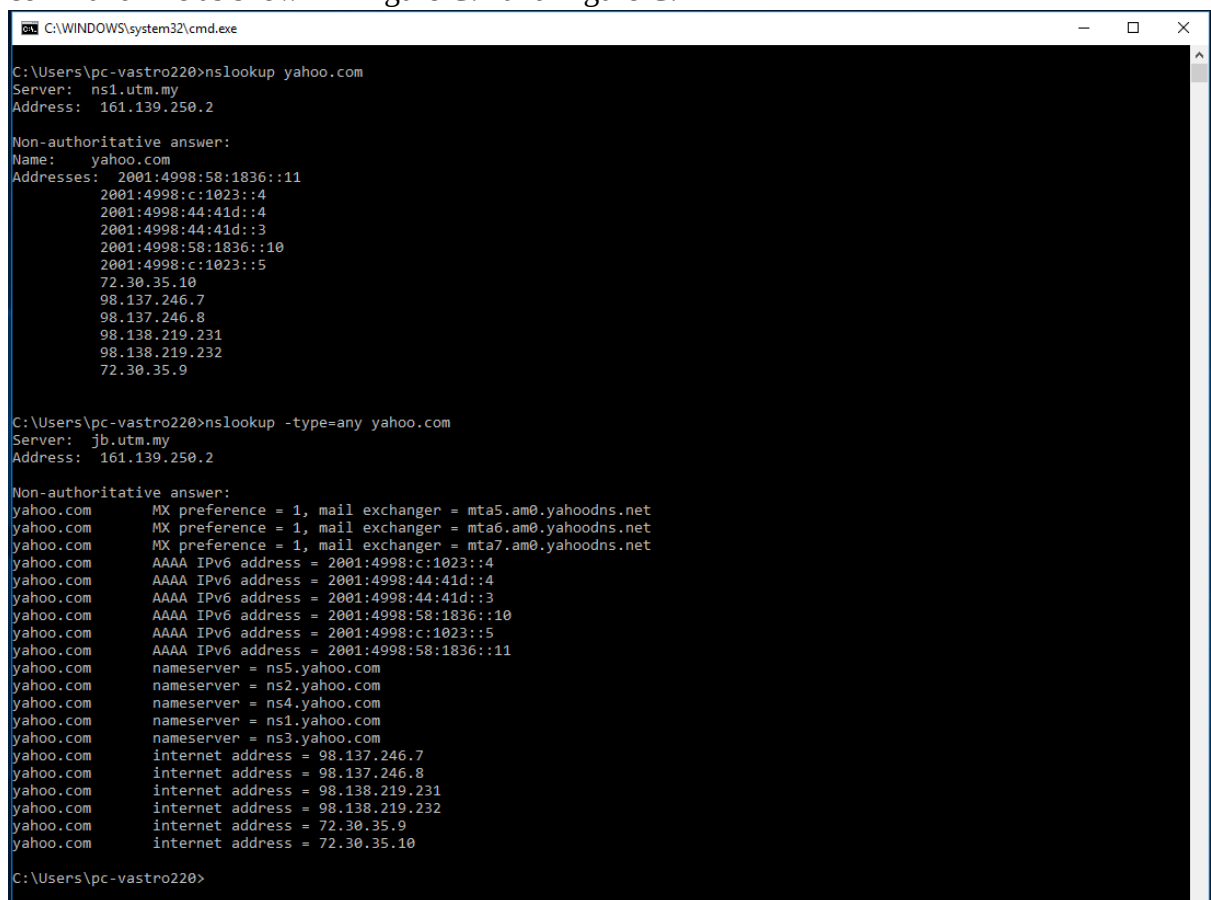


PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

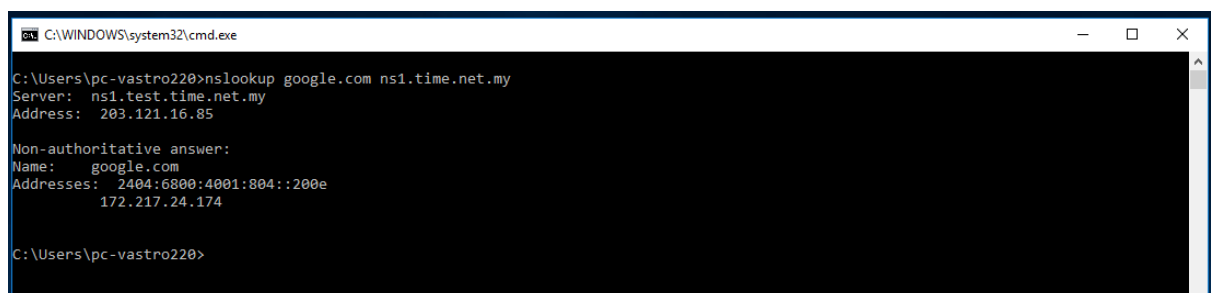
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

IPv4 Address: 104.83.197.169

IPv6 Addresses: 2001:d08:10:989::356e
 2001:d08:10:997::356e
 2001:d08:10:996::356e
 2001:d08:10:999::356e
 2001:d08:10:98c::356e

```
C:\Users\User>nslookup www.microsoft.com
Server:   UnKnown
Address:  192.168.1.254

Non-authoritative answer:
Name:     e13678.dscb.akamaiedge.net
Addresses: 2001:d08:10:989::356e
           2001:d08:10:997::356e
           2001:d08:10:996::356e
           2001:d08:10:999::356e
           2001:d08:10:98c::356e
           104.83.197.169
Aliases:  www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

ns4-39.azure-dns.info

ns3-39.azure-dns.org

ns1-39.azure-dns.com

ns2-39.azure-dns.net

```
C:\Users\User>nslookup -type=ns microsoft.com
Server:   UnKnown
Address:  192.168.1.254

Non-authoritative answer:
microsoft.com    nameserver = ns4-39.azure-dns.info
microsoft.com    nameserver = ns3-39.azure-dns.org
microsoft.com    nameserver = ns1-39.azure-dns.com
microsoft.com    nameserver = ns2-39.azure-dns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all

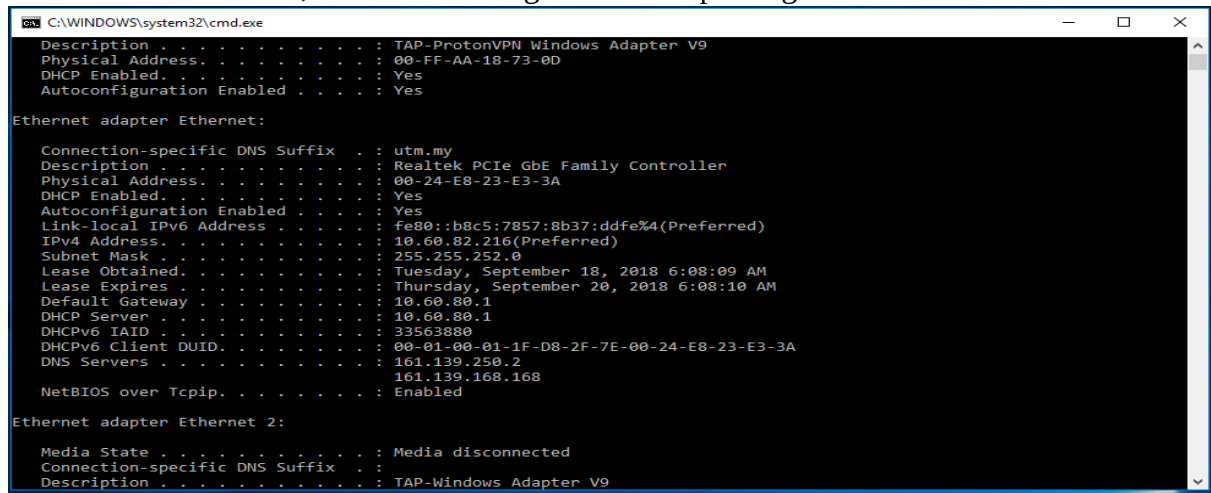


Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds. Command: ipconfig /displaydns

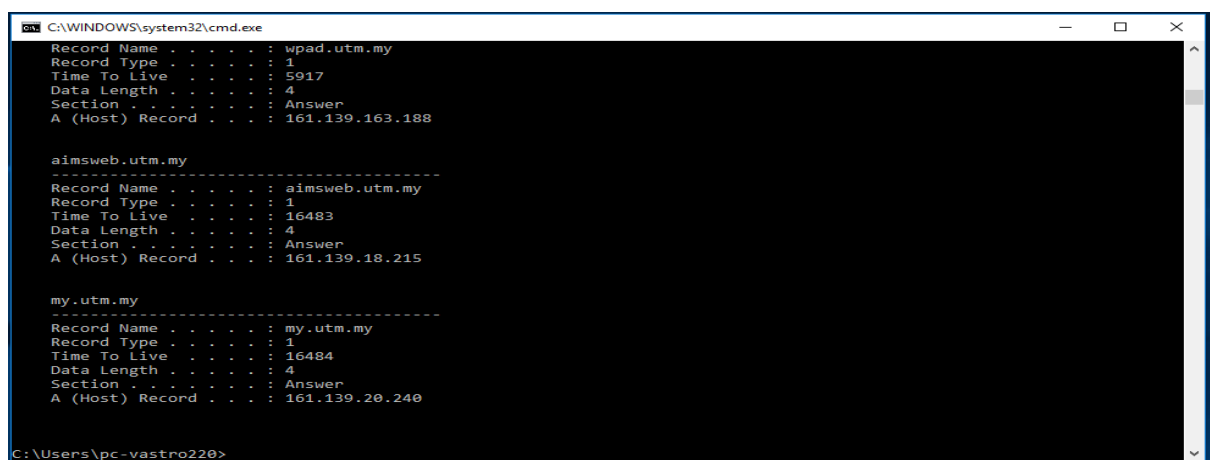


Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



Figure C.5: ipconfig /flushdns result

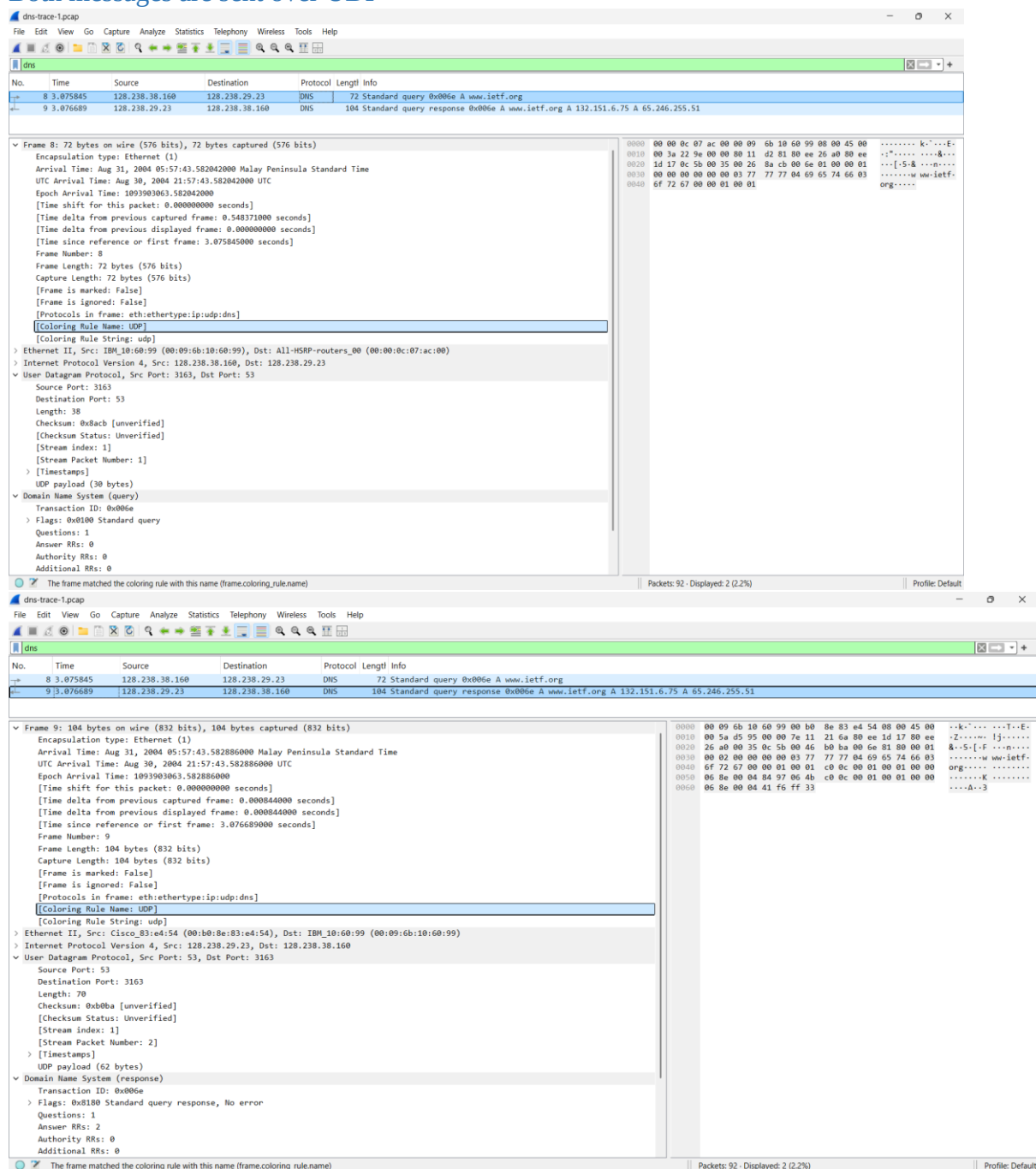
3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1.pcap. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

Frame 8: Query message

Frame 9: Response message

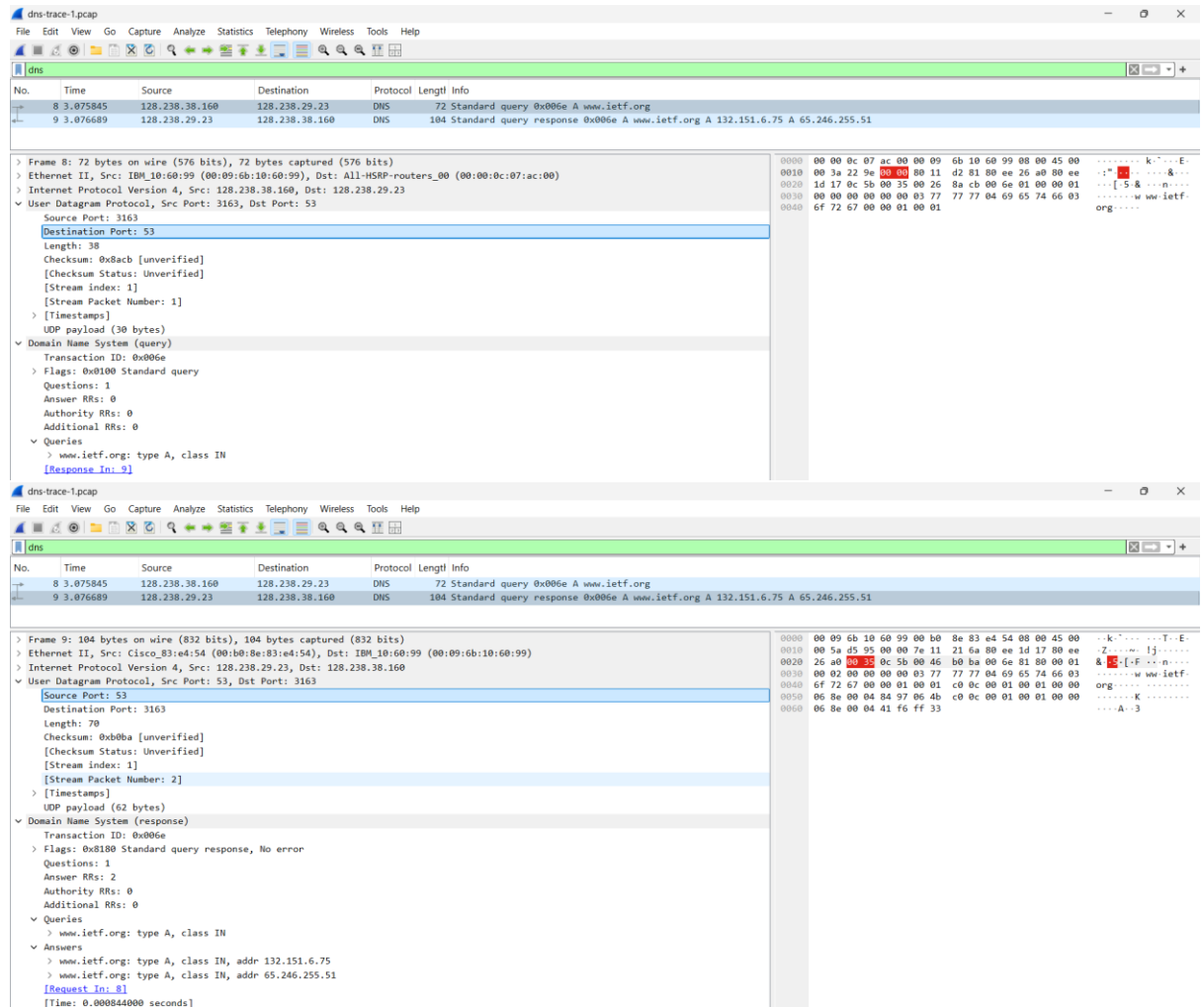
Both messages are sent over UDP



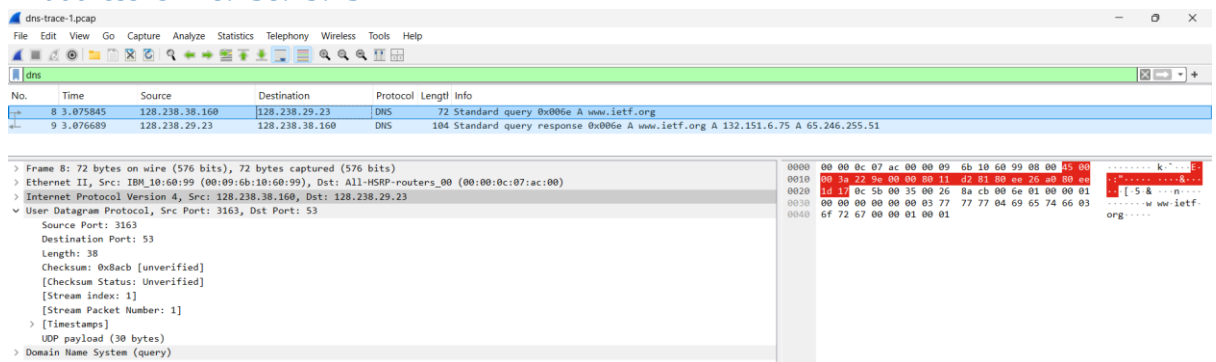
2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port for the DNS query message : 53

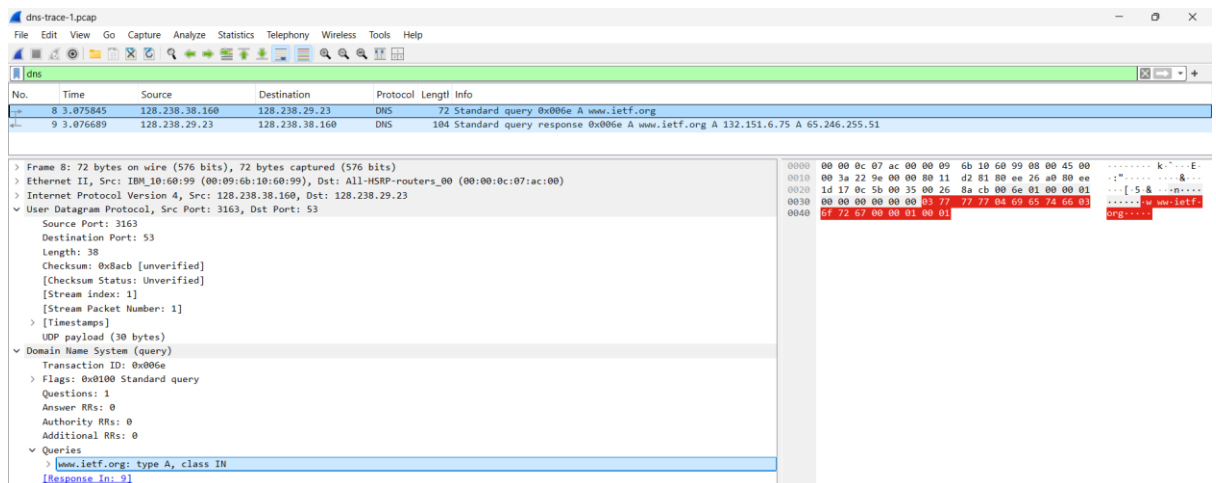
Source port of DNS response message : 53



3. To what IP address is the DNS query message sent? Add screenshots in your answer.
IP address is 128.238.29.23



4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
The DNS query is type A of the domain of www.ietf.org
The query does not contain any “answers”. The Answer RRs shows 0 .



5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.
The DNS response provides two answers, both containing information about ``www.ietf.org``.

Answer 1:

Name: www.ietf.org
Type: A(Host Address)
Class: IN(Internet)
Time to live: 1678 seconds
Data length: 4
Address: 132.151.6.75 (IPv4 address for www.ietf.org)

Answer 2:

Name: www.ietf.org
Type: A (Host Address)
Class: IN (Internet)
Time to live: 1678 seconds
Data length: 4
Address: 65.246.255.51 (IPv4 address for www.ietf.org)

✓ Answers

✓ `www.ietf.org: type A, class IN, addr 132.151.6.75`

Name: `www.ietf.org`
Type: A (1) (Host Address)
Class: IN (0x0001)
Time to live: 1678 (27 minutes, 58 seconds)
Data length: 4
Address: 132.151.6.75

✓ `www.ietf.org: type A, class IN, addr 65.246.255.51`

Name: `www.ietf.org`
Type: A (1) (Host Address)
Class: IN (0x0001)
Time to live: 1678 (27 minutes, 58 seconds)
Data length: 4
Address: 65.246.255.51

[\[Request In: 8\]](#)

[Time: 0.000844000 seconds]

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Yes, the first SYN packet was sent to 132.151.6.75 (the 1st answer's IP address to initiate a new connection between client and server), which corresponds to the first IP address provided in the DNS response message.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=13
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=13
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0

> Frame 10: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)	0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 08 0
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:0	0010 00 30 22 9f 40 00 80 06 a5 b8 80 ee 26 a
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75	0020 06 4b 0d 29 00 50 f2 bd 56 0c 00 00 00 0
> Transmission Control Protocol, Src Port: 3369, Dst Port: 80, Seq: 0, Len: 0	0030 fa f0 ff 7a 00 00 02 04 05 b4 01 01 04 0

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No, there is no new DNS queries before retrieving each image request. After the initial DNS query, the HTTP and TCP continues to use the already resolved IP address (128.238.38.160 and 132.151.6.75) without issue new DNS queries for each image.

This is because DNS queries are typically only made once per domain name within a session or until the Time to Live (TTL) value for the DNS record expires.

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=13
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=13
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2e.gif HTTP/1.1
29	3.192665	132.151.6.75	128.238.38.160	TCP	62	80 → 3371 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0
30	3.192695	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
31	3.192869	128.238.38.160	132.151.6.75	HTTP	314	GET /images/blue.gif HTTP/1.1
32	3.205736	132.151.6.75	128.238.38.160	TCP	60	80 → 3370 [ACK] Seq=1 Ack=267 Win=6432 Len=0
33	3.214651	132.151.6.75	128.238.38.160	TCP	1434	80 → 3370 [ACK] Seq=1 Ack=267 Win=6432 Len=1380
34	3.222185	132.151.6.75	128.238.38.160	TCP	1434	80 → 3370 [ACK] Seq=1381 Ack=267 Win=6432 Len=13
35	3.222249	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=267 Ack=2761 Win=64860 Len=0
36	3.228451	132.151.6.75	128.238.38.160	HTTP	1212	HTTP/1.1 200 OK (GIF89a)
37	3.228509	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=267 Ack=3920 Win=63702 Len=0
38	3.228523	132.151.6.75	128.238.38.160	TCP	60	80 → 3371 [ACK] Seq=1 Ack=261 Win=6432 Len=0
39	3.230578	132.151.6.75	128.238.38.160	HTTP	407	HTTP/1.1 200 OK (GIF89a)
40	3.230629	132.151.6.75	128.238.38.160	TCP	60	80 → 3371 [FIN, ACK] Seq=354 Ack=261 Win=6432 Le
41	3.230667	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=261 Ack=355 Win=64507 Len=0

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port for the DNS query message : 53

Source port of DNS response message : 53

```
> Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-MSRP-routers_00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
> User Datagram Protocol, Src Port: 3742, Dst Port: 53
> Domain Name System (query)
```

```
> Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3742
> Domain Name System (response)
```

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

128.238.29.22

It is not the IP address of my default local DNS server.

The screenshot shows a Wireshark packet capture of DNS traffic. The packet list shows a query from 128.238.38.160 to 128.238.29.22 on port 53. The packet details show the query for www.mit.edu. The packet bytes show the raw data of the query.

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : A0-E7-0B-EF-11-A4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.1.21(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, January 19, 2025 12:05:21 AM
Lease Expires . . . . . : Tuesday, January 21, 2025 12:45:13 AM
Default Gateway . . . . . : 192.168.1.254
DHCP Server . . . . . : 192.168.1.254
DNS Servers . . . . . : 192.168.1.254
NetBIOS over Tcpip. . . . . : Enabled
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

The DNS query is type A of the domain of www.mit.edu

The query does not contain any “answers”. The Answer RRs shows 0.

```
> Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
> User Datagram Protocol, Src Port: 3742, Dst Port: 53
v Domain Name System (query)
  Transaction ID: 0x0003
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
v Queries
  > www.mit.edu: type A, class IN
[Response In: 20]
```

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

1 “answer” is provided.

Answer:

Name: www.mit.edu

Type : A (Host Address)

Class : IN (Internet)

Time to live : 60 seconds

Data length: 4

Address: 18.7.22.83 (IPv4 address for www.mit.edu)

```
v Domain Name System (response)
  Transaction ID: 0x0003
  > Flags: 0x8580 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 3
  Additional RRs: 3
v Queries
  > www.mit.edu: type A, class IN
v Answers
  v www.mit.edu: type A, class IN, addr 18.7.22.83
    Name: www.mit.edu
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
    Address: 18.7.22.83
```